Learning Multiscale Temporal–Spatial–Spectral Features via a Multipath Convolutional LSTM Neural Network for Change Detection With Hyperspectral Images

Changjiang Shi, Zhijie Zhang, Wanchang Zhang, Chuanrong Zhang, and Qiang Xu

***Abstract*— Change detection (CD) with hyperspectral images (HSIs) can be effectively performed using deep learning networks (DLNs) by taking advantage of HSIs for their abundant spectral and spatial information and the excellent performance of DLN in machine learning. By modeling the temporal dependence of multiscale representative features, more discriminative information reflecting land use and land cover (LULC) changes can be obtained by suppressing less correlated information while improving the robustness of pseudo-changes caused by imaging noises. However, preserving time-dependent multiscale representative features while extracting spatial–spectral features based on conventional DLN is difficult, mainly due to the structural limitation of conventional DLN. A multipath convolutional long short-term memory (LSTM) multipath convolutional long short-term memory neural network (MP-ConvLSTM) taking advantage of LSTM and convolutional neural network (CNN) through the designed parallel architecture to learn multilevel temporal dependencies of bitemporal HSIs, therefore, was proposed for extracting multiscale temporal–spatial–spectral features by combining hidden states from different paths of ConvLSTM in the present study. In the proposed MP-ConvLSTM, the efficient channel attention (ECA) module was introduced to refine features of different paths, and Siamese CNN was adopted to reduce HSIs' dimensionality and extract preliminary features to build up an end-to-end trainable model for CD with HSIs. The validity of the MP-ConvLSTM was evaluated using the binary and multiclass CD datasets. The CD accuracy of the proposed MP-ConvLSTM was visually and statistically evaluated by different criteria and compared with those derived from several state-of-the-art (SOTA) CD algorithms. The experiments demonstrated that our proposed model not only outperformed those SOTA CD models but also exhibited better tradeoff between complexity and accuracy in general.**

***Index Terms*— Change detection (CD), efficient channel attention (ECA), hyperspectral images (HSIs), long short-term memory (LSTM), multipath convolutional long short-term memory neural network (MP-ConvLSTM), Siamese convolutional neural network (CNN).**

I. Introduction

CHANGE detection (CD) is one of the important applications of remote sensing and has been extensively applied for various purposes, e.g., land use and land cover (LULC) mapping [1], [2], environmental monitoring [3], [4], disaster management [5], ecosystem assessment [3], and agriculture regional planning [6]. Traditional methods for CD can be subdivided into three categories.

1) Classic image algebraic methods for CD that measures change by calculating the algebraic characteristics between corresponding bands of multitemporal remote sensing images through band differencing [1], band ratio [4], band regression [4], and spectral angle [7]. A significant shortcoming of such method is that it is difficult to determine the change threshold.
2) CD based on image transformation that extracts the image features to infer the changes from the statistical information of the image through mapping spectral information of the image to a specific feature space to distinguish pixels that have changed [8].
3) CD-based on image classification independently to identify the changes from bitemporal images by comparing the LULC types of two images [1].

In the literature, the majority of CD approaches have been proposed for multispectral and synthetic aperture radar images, and only a few works are devoted to CD for hyperspectral images (HSIs) [9]–[15]. Compared with multispectral images, HSIs are typically lower in spatial resolution but much higher in spectral resolution with abundant spectral information [16], which were thought of being capable of identifying subtle changes of LULC [17]. The traditional CD methods with single-band or multispectral images are seldom found in CD with HSIs, mainly due to its high-dimensional features.

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The redundancy of information between adjacent bands and the high dimensionality of HSIs bring extra difficulties in algorithm development for CD with HSIs [18], [19]. For classifications with HSIs, with the increase of bands involved, the classification accuracies usually turn from increasing to decreasing, which is the so-called "Hughes" effect [20], which cannot be ignored in CD with HSIs. Deep neural networks have achieved great success in various computer vision and remote sensing applications in the recent decade [21]. The restricted Boltzmann machine [22], [23] and support vector machine (SVM) [24], [25] were developed at an early stage for CD with HSIs by extracting features from the different maps of bitemporal HSIs. These methods, however, only took the spectral features of pixels but failed to take spatial features into account mainly due to the high dimensionality of HSIs. For reducing the high dimensionality of HSIs, general end-to-end two-dimensional convolution neural network (GETNET) [17] introduced the mixed affinity matrix to 2-D-convolutional neural network (CNN) to excavate more channel gradient features. The mixed affinity matrix dramatically increases the complexity of the model, yet the spatial contextual features are not sufficiently considered. The Siamese network was just proposed as one of the standard methods to account the problems aforementioned [26]–[30]. With Siamese network architecture, a spectral–spatial CNN with Siamese architecture (SSCNN-S) to account similarity of patch pairs in the CD with HSIs was proposed [30]. Spectral-spatial-wise attention (SSA)-SiamNet embeds the convolutional block attention module (CBAM) [31] attention mechanism into the Siamese CNN, emphasizing informative channels and locations and suppressing less informative ones to refine the spectral–spatial features adaptively [28]. The Siamese neural network generates a change intensity map through similarity comparison of features between two bitemporal images and then determines the threshold value of changes after iterative calculations to generate change map from change intensity map, which considerably complicates the process of CD. The 3-D-CNN takes some advantages of 2-D-CNN in comprehensive learning on spatial–spectral features for purpose of CD [32], [33]. Simplified 3-D convolutional autoencoder (S3DCAECD) [34] combines a deep unsupervised autoencoder with a 3-D convolutional network to extract deep spectral–spatial features while reducing spectral redundancy. Three-directions spectral-spatial convolution neural network (TDSSC) [35] was proposed for 3-D decomposition of hyperspectral change tensors by the 3-D CNN; however, the complexity of the 3-D CNN is high, and the method is sensitive to a batch size of datasets and the number of input bands.

Most CD algorithms adopt relatively simple operations to bitemporal images such as image differencing or stacking but ignore the time dependency among multitemporal images. In the CD task, bitemporal images as input are acquired at different times. The pixels in the images corresponding to fine spectral profiles are of high spectral resolution. The pair from the bitemporal spectral profiles of the changed and unchanged pixels has different magnitudes and spectral ranges of irradiance changes (see Fig. 1). The magnitude of irradiance change of the pixels is usually greater before and after the LULC change. Moreover, spectral profiles of those pixels that experienced LULC changes exhibit a larger spectral range than those without LULC changes. Theoretically, the spectral profiles of the bitemporal pixels should be the same for pixels without LULC change, and the band irradiance response at the same pixel position in both images depends only on the LULC change. In other words, the basic premise by using remotely sensed data for CD is that changes in the objects of interest will result in changes in reflectance or local textures that are separable from changes caused by other factors such as differences in atmospheric conditions, illumination, and viewing angles at bitemporal images acquired. However, the above premise is difficult to satisfy in practice, HSIs sometimes are corrupted by strip noise, Gaussian white noise, and observation conditions (atmospheric and radiation conditions), and even pixels without LULC changes may have significant irradiance and spatial textural changes in some bands. To focus on the CD task of LULC, strengthening those time-dependent properties associated with LULC changes but suppressing the pseudo-changes caused by acquisition conditions and noise to derive representative spatial–spectral feature responses of LULC changes is vital for fulfilling the task. Recurrent neural network (RNN) [36] takes an obvious advantage over CNN in time-dependency modeling [37], [38]. The ReCNN [39] architecture combines a CNN and an RNN to form an end-to-end network for not only generating spectral–spatial features but also analyzing the time-dependent of two multispectral images. However, these networks are not sufficiently adaptive for HSIs of high dimensionality and information redundancy between adjacent channels. The Re3FCN [40] utilizes a recurrent 3-D fully convolutional network (FCN) that extracts the spectral–spatial features from the HSIs simultaneously. Compared with 2-D CNNs, 3-D CNNs have more parameters and take longer time for training. Moreover, the gradient attenuation effect of long short-term memory (LSTM) in the training process cannot be ignored [41].

In addition, it should be noted that the above study does not consider the scale effect of the channel when modeling the temporal dependence, and once the LULC corresponding to a pixel is changed, the radiance change of that pixel includes several continuous spectral ranges, rather than a few individual channels alone. The hierarchical structure proposed by hierarchical multiscale (HM)-RNN [42] can model different scales of temporal dependencies, with low update frequency at higher levels, and high update frequency at lower levels, therefore, takes the advantage of capturing patterns at different channel granularities, with higher levels capturing changes in spectral trends and lower levels being more sensitive to local changes in spectra. Inspired by HM-RNN that can capture the latent hierarchical structure in the sequence by encoding the temporal dependencies with hierarchical structures, we consider introducing hierarchical and skip connection structures to RNNs for enhancing representativeness of multiscale channels for the CD with HSIs.

To reduce the impacts of: 1) high dimensionality, adjacent band redundancy; 2) pseudo-changes caused by acquisition conditions and noise; and 3) variability of channel scale on HSI CD task, the present study proposed a novel deep learning network (DLN) for CD with HSIs, entitled the

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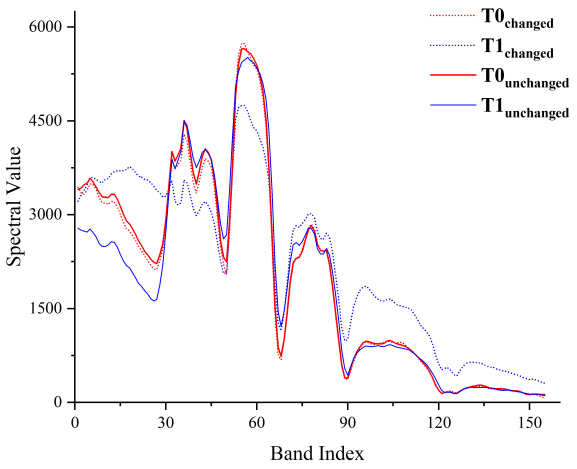

Fig. 1. Spectral profiles of a changed (dashed line) and an unchanged (solid line) pixel from bitemporal HSIs.

multipath convolutional long short-term memory neural network (MP-ConvLSTM), which took the advantages of LSTM and CNN through the skillfully designed parallel architecture that enables the DLN to learn multilevel time dependencies of bitemporal HSIs for extracting joint multiscale temporal–spatial–spectral features by combining hidden states from different paths of ConvLSTM block for final CD. In the proposed MP-ConvLSTM, the efficient channel attention (ECA) module was introduced to refine the most representative features of different paths, and we adopted Siamese CNN (Siam CNN) for reducing HSI dimensionality and extracting preliminary features to build up an end-to-end trainable model for the CD with HSIs. The main functions of the major models integrated in the proposed method are summarized as follows.

1) The Siamese CNN was adopted to extract spatial–spectral features and to reduce the dimensionality of the input HSIs. In the CNN feature extraction process, the feature map of each layer was the summation of multiple feature maps of the previous layer through 2-D convolution, and the final feature map was obtained by weighting 2-D images of different bands. Therefore, the feature map represented both spatial and spectral dimensional information. The dimensionality reduction was achieved by decreasing the number of convolution kernels. Since most patch pairs have comparable feature properties before and after the LULC changes, feature extraction with a Siamese CNN just shared weights of the two network branches of the Siamese structure reasonably.
2) The temporal dependencies between two HSIs acquired at different times were expressed as the response of the reflectance from different bands to changes. To strengthen time-dependent properties associated with LULC changes, meanwhile suppress the pseudo-changes caused by acquisition conditions and noise, and derive representative spatial–spectral features that responded to LULC changes, the ConvLSTM was introduced to model temporal change dependencies of bitemporal HSIs while preserving spatial information.
3) MP-ConvLSTM took the advantages of LSTM and CNN through the skillfully designed parallel architecture that enables the DLN to learn multilevel time dependencies of bitemporal HSIs for extracting joint multiscale temporal–spatial–spectral features by combining hidden states at different depths from MP-ConvLSTM through the skip connection and the ECA module for final CD.

The rest of this article consists of the following sections. Section II describes the problem addressed in this article, and the details of the proposed MP-ConvLSTM are presented. Section III evaluates the performances of the proposed MP-ConvLSTM using four different remotely sensed benchmark datasets and compares them with state-of-the-art (SOTA) CD approaches. The experiments for assessing the performance of the proposed MP-ConvLSTM in CD were then presented and discussed in Section IV, followed by a discussion and conclusion presented in Section V at the end.

II. Methodology

This study establishes an end-to-end neural network model for HSI CD based on patches. For the binary CD task, utilizing the change vector analysis (CVA) method with high confidence threshold to generate the training set with labels according to the strategy of training with predetection [17], [43]. Also, for the multiclass CD task, we randomly selected pixels as the training set. Bitemporal patches were extracted from the training sample pixel and its surrounding pixels and then were fed into the neural network to train the CD model. The output of the model is the change probability of the center pixel of the patch, thus ensuring pixel-by-pixel CD.

An architectural overview of the proposed MP-ConvLSTM is shown in Fig. 2. First, preliminary spatial–spectral feature extraction and dimensionality reduction were performed on the input patch pairs of HSIs simultaneously by the Siamese CNN for binary CD or principal component analysis (PCA) for multiclass CD. Second, the spatial–spectral feature maps generated by Siamese CNN were fed into the MP-ConvLSTM block to learn the temporal dependencies at multiscale (in channel dimensions) among bitemporal spatial–spectral feature maps to enhance the channels that are sensitive to changes and obtain the joint multiscale representative temporal–spatial–spectral features. In this process, the ECA module fuses the features from different levels. Finally, the fused features are passed through the convolution and pooling layers to get the change probabilities of the central pixel of the HSI patch. The detail of three modules, namely, Siamese CNN, MP-ConvLSTM block, and ECA module, how was integrated into the proposed MP-ConvLSTM will be introduced in Section II-A–II-C, respectively.

A. Siamese CNN

For binary CD task, the Siamese architecture is designed according to the following considerations. First, the traditional CD method combines the HSIs of bitemporal along the spectral dimension or takes the difference of two images as one HSI for processing, which causes the loss of temporal information. In contrast, the Siamese CNN includes two branches, and the output is still a pair of feature tensors after

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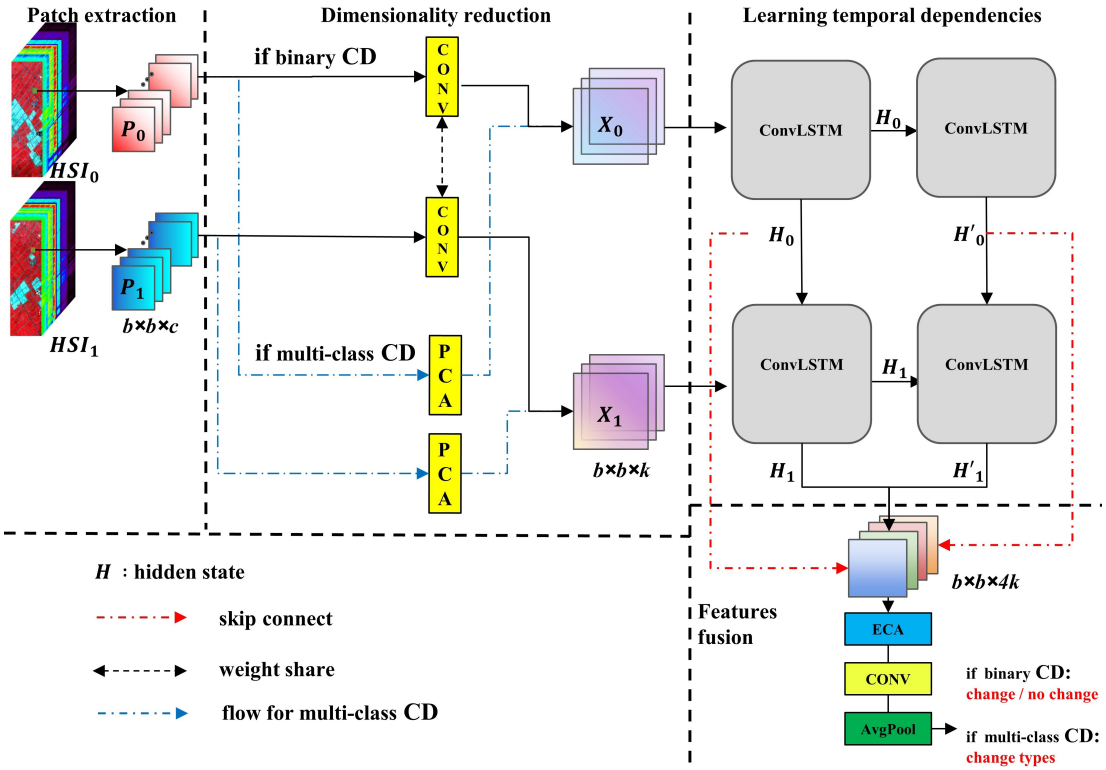

Fig. 2. Architecture of the proposed multipath convolutional LSTM for CD.

processing the original image pair. Siamese CNN can learn trainable nonlinear functions to map patches from the bitemporal HSI patches to tensors in feature space so that the distance (e.g., Euclidean distance) between two vectors is small if the two patches are similar and large when they changed. Also, to alleviate the requirements of the high-dimensional bands of HSIs for computer memory and arithmetic power, the spectral dimension needs to be reduced prior to perform time-dependent feature extraction. Therefore, in the present study, a Siamese CNN network, including two branches of 2-D convolutional layers (with the number of convolutional kernels less than the number of input image channels), was used to extract preliminary features, meanwhile reducing the spectral dimensionality of raw HSIs.

However, it should be noted that the above analysis is only for binary CD. The Siamese structure is suitable for detection of a specific change type rather than for the multiclass CD task. PCA method, therefore, was employed to reduce dimensions of bitemporal feature of HSIs in multiclass CD tasks, which will be analyzed further in Section IV-B. Let $\text{HSI}_t \in \mathbb{R}^{h \times w \times c}$ represent the HSI at time t, where h, w, and c denote the height, width, and number of bands of the HSI, respectively. The patches $P_t^{ij} \in \mathbb{R}^{b \times b \times c}$ are constructed by extracting the pixel at the location (i, j) of $\text{HSI}_{t \in \{0,1\}}$ and its surrounding pixels with size $b \times b \times c$, where b represents the spatial size. The spectral dimension of the patch is reduced to k after dimensionality reduction, and the spatial dimension holds unchanged. To facilitate the following presentation, the reduced bitemporal patches are denoted by X_0 and $X_1 \in \mathbb{R}^{b \times b \times k}$.

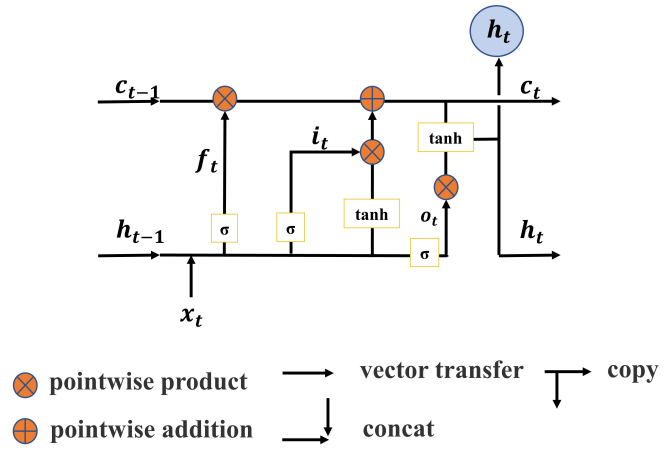

Fig. 3. Structure of the LSTM unit.

B. Multipath ConvLSTM Block

To learn the temporal dependencies between bitemporal patches while performing spatial–spectral feature extraction, the ConvLSTM network was introduced. The ConvLSTM combines the LSTM temporal dependence modeling and the CNN spatial–spectral feature extraction capability. CNNs and other feedforward neural networks commonly used in computer vision and image processing can learn rich spatial–spectral features in images, but prerequisite is that all inputs are independent of each other [36].

In the CD task, radiation change at some wavelengths among bitemporal HSIs is not induced by the LULC change

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but is caused by the noises. Emphasizing bands that can detect changes while suppressing bands that may produce pseudo-changes is vital for relieving noise's influence. Therefore, enhancing time dependencies associated with LULC changes and meanwhile suppressing pseudo-changes caused by acquisition conditions and noise to derive a representative response of spatial–spectral features to LULC changes are essential for CD with HSIs. Unlike CNNs, RNNs are specifically developed to address dependent sequential inputs and solve sequential learning problems. RNNs can explore the temporal dependencies of multitemporal images by using recursive implicit states to link the previous image to the current one. LSTM [44] is a classical RNN architecture, which is well suited for recognition, handling, and forecasting based on time series of data. An LSTM unit, as shown in Fig. 3, consists of a cell c_t, an input gate i_t, an output gate o_t, a forget gate f_t, an input data x_t, and a hidden state h_t also known as output vector of the LSTM unit. The LSTM cell memorizes values at arbitrary time intervals, and the three gates (forget gate, input gate, and output gate) regulate the inflow and outflow of information at time steps t, which can be calculated as follows:

$$\begin{aligned} i_t &= \sigma(W_{xi}x_t + W_{hi}h_{t-1} + W_{ci} \circ c_{t-1} + b_i) \\ f_t &= \sigma\left(W_{xf}x_t + W_{hf}h_{t-1} + W_{cf} \circ c_{t-1} + b_f\right) \\ c_t &= f_t \circ c_{t-1} + i_t \circ \tanh(W_{xc}x_t + W_{hc}h_{t-1} + b_c) \\ o_t &= \sigma(W_{xo}x_t + W_{ho}h_{t-1} + W_{co} \circ c_t + b_o) \\ h_t &= o_t \circ \tanh(c_t) \end{aligned} \tag{1}$$

where W, b, σ, and $\circ$ denote coefficient matrix, bias vector, sigmoid function, and Hadamard product, respectively. The traditional LSTM is suitable for 1-D signal processing and modeling. Its structure cannot effectively utilize the contextual information of pixels and thus is not ideal for 2-D signal processing such as images and feature maps. ConvLSTM is an extension of LSTM with convolutional operations during the state-to-state transitions. The combination of multiple ConvLSTMs can establish an encoding-prediction structure suitable for extracting temporal–spatial–spectral features from bitemporal HSIs.

Due to the introduction of the convolutional structure, all inputs X_t, cell states C_t, hidden states H_t, and gates (I_t, F_t, and O_t) are 3-D tensors $\in \mathbb{R}^{b\times b\times k}$, the first two dimensions encode spatial feature, and the last dimension learns spectral feature representations. Then, the temporal dependencies were obtained by constructing time series for H and C, as shown in Fig. 4. The input X_t and past states C_{t-1} and H_{t-1} are inputs to the ConvLSTM for determining the future states C_t and H_t, and the critical equations of the ConvLSTM can be summarized as follows according to [45]:

$$\begin{aligned} I_t &= \sigma(W_{XZ} * X_t + W_{HI} * H_{t-1} + W_{CI} \circ C_{t-1} + b_I) \\ F_t &= \sigma(W_{XF} * X_t + W_{HF} * H_{t-1} + W_{CF} \circ C_{t-1} + b_F) \\ C_t &= F_t \circ C_{t-1} + I_t \circ \tanh(W_{XC} * X_t + W_{HC} * H_{t-1} + b_C) \\ O_t &= \sigma(W_{XO} * X_t + W_{HO} * H_{t-1} + W_{CO} \circ C_t + b_0) \\ H_t &= O_t \circ \tanh(C_t) \end{aligned} \tag{2}$$

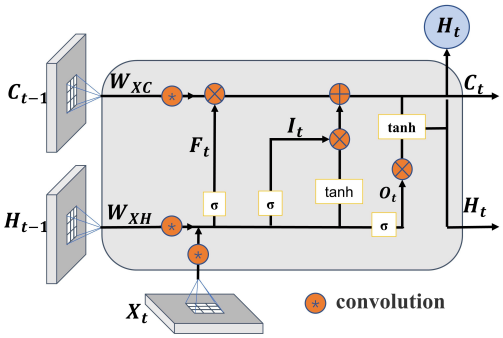

Fig. 4. Inner structure of ConvLSTM.

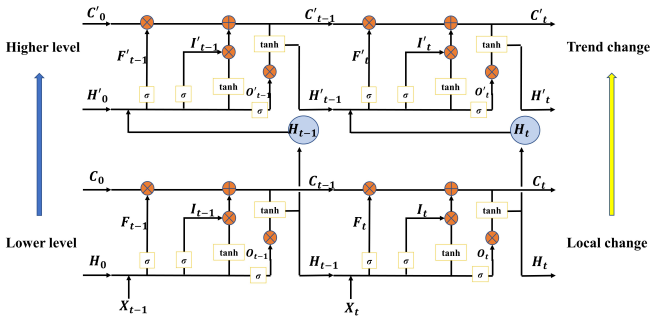

Fig. 5. Parallel block of the proposed MP-ConvLSTM.

where $*$ denotes the convolution operator and $\circ$, as aforementioned, denotes the Hadamard product.

By the convolution operation, the number of parameters is reduced by weight sharing on the one side, and the spatial domain information is effectively used by taking spatial neighborhood pixels into account on the other side. The hidden states H_t are deemed as short-term memories in the time domain, while the cell states C_t are considered to be long-term memories. For the CD task, the input data are HSIs acquired at two different times, which are more concerned with short-term time dependence that emphasizes bands capable of detecting changes and suppressing bands of less information. ConvLSTM recognizes and analyzes the multitemporal changes with HSIs by modeling temporal dependencies to take temporal features into account in CD. Hence, the hidden states H_t of the ConvLSTM output can be extracted as representative features of changes.

However, the LSTM was originally designed for natural language processing tasks with long sequences but a small number of features per sequence, while in the HSI CD task, there are only two timesteps but has a larger number of channels per sequence. For this reason, a parallel structure, instead of simply stacking the ConvLSTM layers in series to effectively extract multiscale features in the channel dimension and describe more complex input patterns, is presented in Fig. 5. The low-level hidden states are again fed into ConvLSTM to obtain high-level hidden states for extracting short-term dependencies at multiple levels [46]. Higher levels capture changes in spectral trends, and lower levels are more

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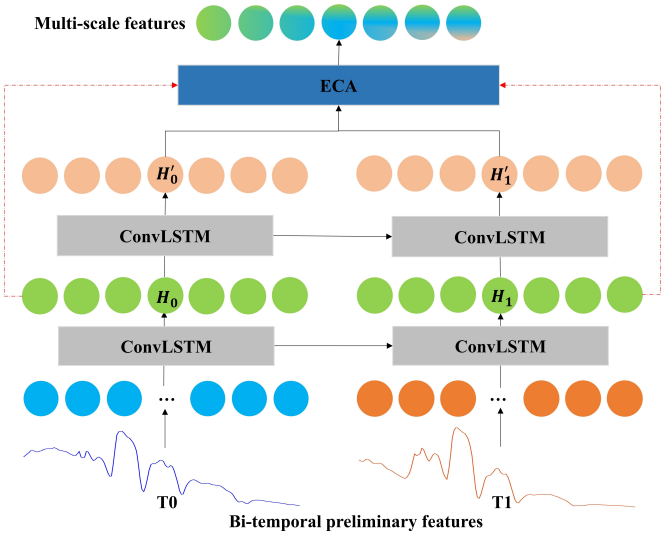

Fig. 6. Extraction of multiscale features in the channel dimension.

sensitive to local changes in spectra. Moreover, to alleviate the possible gradient disappearance caused by layer deepening, concatenating the output of the hidden state at the lower level with the hidden states at the high level as the final output of the entire block is shown in Fig. 6 so that the network enables multilevel modeling of short-term dependence by encoding the temporal dependencies with the network's parallel structure to strengthen the learning of discriminative temporal–spatial–spectral features. As a result, the bitemporal feature tensor X_0 and X_1 outputted from the Siamese CNN or PCA, and after features extraction by MP-ConvLSTM block, the multiscale temporal–spatial–spectral feature representation tensor H_0, H_0', H_1, and H_1' thus can be derived.

C. ECA Module

Although the outputted feature maps of the MP-ConvLSTM (H_0, H_0', H_1, and H_1') are in the same size, the scale of spatial–spectral features and temporal dependencies varies in certain levels. An automatic channel selecting mechanism is required to restrain redundancy of the channel-to-channel information when fusing these features. The attention mechanism proposed by the squeeze and excite network (SENet) [47] can recalibrate channel-wise feature reactions to model the mutual dependencies across channels, allowing the model to focus more on the most informative channel features but suppress those unimportant ones, and the fundamental equation is defined as

$$M_{\text{se}}(F) = \sigma\left(\text{MLP}(\text{AvgPool}(F)) + \text{MLP}(\text{Max})\text{Pool}(F)\right) \tag{3}$$

where F is the input feature map, M_{se} denotes the attention map, and multilayer perception (MLP), AvgPool, and MaxPool represent MLP, average-pooling, and max-pooling operations, respectively. The attention mechanism of SENet first performs channel compression on the input feature map, and such compression and dimensionality reduction has a detrimental effect on the learning of dependencies between channels.

To avoid channel compression, the ECA [48] uses a locally spanning channel interaction strategy, which is effectively implemented by the 1-D convolution, preserving good performance of the model while significantly decreasing model complexity. The critical equation of the ECA module is given as follows:

$$F = \left[H_0, H_0', H_1, H_1'\right] \tag{4}$$

$$M_{\text{eca}}(F) = \sigma\left(C1D_k(F)\right) \tag{5}$$

$$F_{\text{eca}} = M_{\text{eca}} \otimes F \tag{6}$$

where $C1D$ denotes the 1-D convolution, $\otimes$ denotes the element-wise product, and k is the kernel size of 1-D convolution and was set to 3 in the experiments. $M_{\text{eca}} \in \mathbb{R}^{1\times1\times4k}$ denotes the ECA attention map, and F is stack of different semantic-level features (H_0, H_0', H_1, and H_1') from MP-ConvLSTM and can be refined as the fused features F_{eca} input to the final convolution layer, average-pooling layer to derive central pixel's change probability of the patch. Pixels with a high probability of changes are identified as the changed pixels, and pixels with a low probability of changes are identified as unchanged.

III. Experiments

A. Description of the HSIs Used for Experiments

Four public assessable HSIs datasets acquired by the Earth Observing-1 (EO-1) Hyperion were used to evaluate the effectiveness and reliability of the proposed method. The EO-1 Hyperion covers the spectral range from 0.4 to 2.5 μm with 242 spectral bands. In addition, its spectral resolution is about 10 nm with a spatial resolution of 30 m. For the binary CD task, the sample generation method is referred to [17], [43]. The present study utilizes the traditional CVA method with high confidence threshold to generate pseudo training datasets with labels. Pixels that were highly likely to be identified correctly were selected as final labeled samples to train the network. The number of training and testing samples for the four experimental HSIs is summarized in Table I. The details of these datasets were presented as follows.

The first dataset of HSIs for CD, entitled as "Farmland," contains images of a farmland area in Yancheng City, Jiangsu Province, China. The two HSIs in the dataset were acquired on May 3, 2006, and April 23, 2007. The size of the spatial dimension of this dataset is 450 $\times$ 140, and the number of spectral bands is 155 after preprocessing. The cultivated land changes represent the main changes in this dataset, and other types of changes are also labeled. Referring to the setting in [17], we randomly selected 4400 changed pixels and 8800 unchanged pixels, approximately accounting for 20.95% of the total pixels for training the model, and the remaining pixels were used for testing the model.

The second HSIs dataset for CD, "River," covers a region in Jiangsu province, China. Two HSIs were acquired on May 3, 2013 and December 31, 2013. The HSIs dataset contains 198 bands with 463 $\times$ 241 pixels after excluding the noise bands. The LULC changes mainly take place over riverside sediment, and other types of LULC changes were also labeled. For detail, please refer to the settings in [17]. About 3.37%

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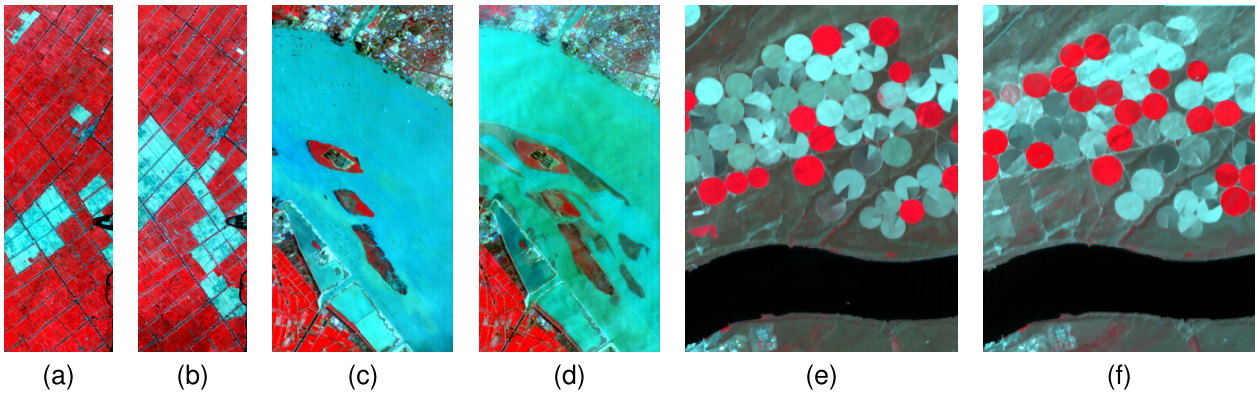

Fig. 7. Visualization of the HSIs utilized for binary CD experiments. (a) "Farmland" HSIs acquired on May 3, 2006. (b) "Farmland" HSIs acquired on April 23, 2007. (c) "River" HSIs acquired on May 3, 2013. (d) "River" HSIs acquired on December 31, 2013. (e) "USA" HSIs acquired on May 1, 2004. (f) "USA" HSIs acquired on May 8, 2007.

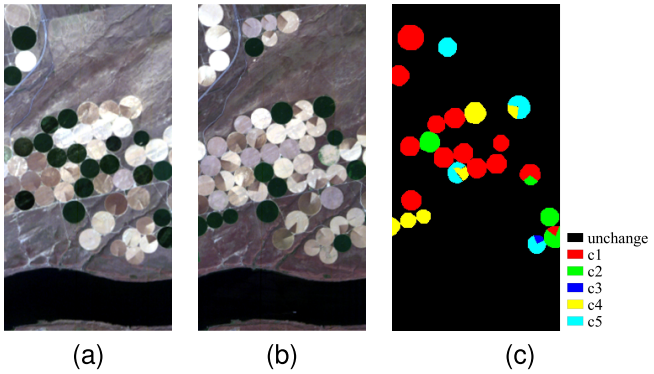

Fig. 8. Visualization of the HSIs and ground truth utilized for multiclass CD experiments. (a) "Hermiston" HSIs acquired in 2004. (b) "Hermiston" HSIs acquired in 2007. (c) Ground truth map of five types of changes related to crop transitions.

of the total pixels of this dataset were randomly selected as the training dataset, of which 1250 are the changed pixels and 2500 are the unchanged pixels. The remaining pixels were used for testing.

The third HSIs dataset for CD, "USA," was acquired from the irrigated farmland on May 1, 2004, and May 8, 2007, where LULC types of bared soil, the irrigated land, and river take the dominant part in Hermiston, Umatilla County, Oregon, USA, and all the changed pixels for each LULC types were labeled. This dataset is composed of 154 bands with a spatial size of 307 × 241 pixels. LULC changes mainly take place on irrigated farmland and rivers. Referring to the settings in [49], we randomly selected approximately 9.77% of total pixels for training, of which 3313 were the changed pixels and 3919 were the unchanged pixels. The remaining pixels were used for testing. Fig. 7 presents the false-color composites [bands 33, 22, and 11 as red green blue (RGB)] of the above three HSIs datasets for binary CD experiments.

The fourth HSIs dataset, "Hermiston," was used for multiclass CD, which covers the crop fields in Hermiston, Umatilla County, OR, USA, in 2004 and 2007. Unlike the three datasets mentioned above, the "Hermiston" dataset labeled five types of changes related to pixel-level crop transitions. This dataset

TABLE I

NUMBER OF TRAINING AND TESTING SAMPLES FOR THE THREE BINARY CD DATASETS

Dataset	Training Set			Testing Set		
	Unchanged	Changed	Total	Unchanged	Changed	Total
Farmland	8800	4400	13200	35923	13877	49800
River	1250	2500	3750	99385	8448	107833
USA	3919	3313	7232	53392	13363	66755

TABLE II

NUMBER OF TRAINING AND TESTING SAMPLES OF "HERMISTON" DATASET FOR THE MULTICLASS CD

Change Types	Training Set	Testing Set	Total
unchange	12000	56014	68014
c1	1000	4558	5558
c2	200	1131	1331
c3	20	59	79
c4	300	1257	1557
c5	300	1161	1461

is composed of 242 bands with the spatial size of 390 × 200 pixels. We randomly selected approximately 17.71% of total pixels for training, of which 1820 were the changed pixels and 12 000 were the unchanged pixels. The distribution of the types is listed in Table II. The remaining pixels are for testing. Fig. 8 presents the false-color composites (bands 33, 22, and 11 as RGB) of the HSI dataset for multiclass CD experiments.

B. Evaluation Measures

To evaluate the aptness of our proposed method, we compared the obtained CD maps with the reference maps, where white pixels correspond to the changed parts and black pixels correspond to the unchanged parts. The accuracy of the results can directly reflect the reliability and performance of the proposed MP-ConvLSTM. The present study selected five pixel-level evaluation criteria for examining the performance of the proposed MP-ConvLSTM: overall accuracy (OA), kappa coefficient (kappa), precision, recall, and F1-score (F1). There

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are four indices to determine OA, kappa coefficient, precision, recall, and F1:

1) true positive (TP), i.e., the number of the changed pixels that is correctly detected;
2) true negative (TN), i.e., the number of the unchanged pixels that is correctly detected;
3) false positive (FP), i.e., the number of pixels with false alarms;
4) false negative (FN), i.e., the number of the changed pixels is incorrectly detected.

The OA is defined as

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}. \tag{7}$$

The kappa coefficient is applied as a consistency measure and is a metric for assessing classification accuracy. In the CD task, the kappa coefficient represents the agreement between the predicted CD map and the referred CD map. The kappa coefficient is calculated as

$$\mathrm{Kappa} = \frac{\mathrm{OA} - P}{1 - P} \tag{8}$$

$$\mathrm{OP} = \frac{(\mathrm{TP} + \mathrm{FP})(\mathrm{TP} + \mathrm{FN})}{(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN})^2} + \frac{(\mathrm{FN} + \mathrm{TN})(\mathrm{FP} + \mathrm{TN})}{(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN})^2}. \tag{9}$$

Precision measures the proportion of pixels identified as changed that are changed in real and can be calculated as

$$\mathrm{precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{10}$$

Recall measures the proportion of the changed pixels that identified correctly as the changed and can be estimated as

$$\mathrm{recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}. \tag{11}$$

F1-score is the harmonic average of precision and recall and can be calculated by (12). F1-score combines the results of precision and recall, and when F1 is high, it indicates that the CD method is more effective

$$\mathrm{F1}_{\mathrm{score}} = \frac{2 * \mathrm{precision} * \mathrm{recall}}{\mathrm{precision} + \mathrm{recall}}. \tag{12}$$

C. Experimental Results

To evaluate the validity of the proposed MP-ConvLSTM model in terms of CD task, we ran it ten times on each dataset and obtained the mean value together with the standard deviation of the OA, kappa coefficients, precision, recall, and F1 as evaluation metrics for CD. Other SOTA methods, including CVA [50], PCA-CVA [51], SVM [25], patch-based CNN (PBCNN) [52], 3-D-CNN [33], LSTM [53], LSTM-FCN [54], GETNET [17], SSCNN-S [30], and SSA-SiamNet [28], were reimplemented for comparisons. As the CVA and PCA-CVA methods are only applicable to binary CD, we did not conduct experiments with these two methods for the "Hermiston" dataset.

All the experiments in this study were carried out on a Windows personal computer (PC) with a GeForce RTX 2080Ti graphics processing unit (GPU) and an Intel i9-10900K central processing unit (CPU). The model of MP-ConvLSTM was implemented by the PyTorch framework. With all deep learning-based methods during the training and testing phase, the batch size was set to 16, and the patch size and reduction rate were set to 3 × 3 and 16 for all four experimental datasets, respectively. Moreover, Adam was employed as an optimizer, and cross-entropy loss was chosen as the function for model optimization. The l2 regularization coefficient was set to $5e^{-4}$. The learning rate scheduler was cosine annealed warm restart. Every 50 epochs, the learning rate decreases from a maximum of $1e^{-4}$ to a minimum of $1e^{-5}$ and then starts back to $1e^{-4}$ again, and the model reached convergence at around 200 epochs.

For sake of comparison, we compared the kappa coefficients of different methods for different datasets through the bar chart in Fig. 9. Figs. 11–14 present the CD maps obtained from four experimental datasets with different models selected for comparisons. Table III lists quantitative accuracy comparisons for different models. For evaluating the tradeoff between the complexity and accuracy of each model, the number of trainable parameters and the floating point of operations (Flops) of different models are also provided and listed in Table IV with the four experimental HSIs for reference.

TABLE III
QUANTITATIVE ANALYSIS OF CD METHODS FOR THE FOUR DATASETS

Dataset	Method	OA(%)	Kappa(%)	Precision(%)	Recall(%)	F1(%)
Farmland	CVA [50]	95.31	88.47	90.13	93.41	91.74
	PCA-CVA [51]	96.49	91.46	90.85	97.20	93.92
	SVM [25]	97.20±0.10	93.13±0.24	94.16±0.16	95.85±0.26	94.98±0.18
	GETNET [17]	97.79±0.02	94.46±0.05	94.39±0.11	97.74±0.05	95.99±0.03
	PBCNN [52]	97.61±0.05	94.13±0.05	95.01±0.29	96.59±0.34	95.79±0.04
	3D-CNN [33]	98.01±0.09	95.20±0.28	94.79±0.04	**98.33±0.31**	96.54±0.17
	SSCNN-S [30]	98.45±0.07	96.13±0.06	96.96±0.20	97.44±0.23	97.19±0.04
	SSA-SiamNet [28]	98.47±0.11	96.36±0.12	97.23±0.06	98.14±0.12	97.41±0.08
	LSTM [53]	97.14±0.10	93.09±0.25	94.87±0.26	95.11±0.28	94.97±0.18
	LSTM-FCN [54]	97.42±0.02	93.51±0.06	97.14±0.23	93.78±0.30	95.34±0.05
	MP-ConvLSTM (Ours)	**98.89±0.06**	**97.23±0.15**	**97.80±0.18**	98.20±0.27	**98.00±0.11**
River	CVA	92.73	63.75	51.93	**96.15**	67.44
	PCA-CVA	92.61	63.26	51.53	95.75	67.00
	SVM	94.73±0.20	68.58±1.15	62.91±1.69	82.38±0.20	71.30±1.04
	GETNET	95.51±0.11	74.42±0.24	65.34±2.56	92.84±2.75	76.76±0.20
	PBCNN	96.02±0.12	71.61±0.47	75.11±2.21	72.62±1.96	73.79±0.41
	3D-CNN	96.71±0.09	73.76±0.38	**95.31±1.24**	62.21±0.81	75.39±0.34
	SSCNN-S	95.54±0.18	74.08±0.26	64.95±1.07	93.25±1.77	76.51±0.25
	SSA-SiamNet	97.35±0.08	82.01±0.41	82.26±1.59	84.50±2.05	83.36±0.38
	LSTM	95.58±0.09	73.08±0.74	67.32±0.31	85.51±1.51	75.45±0.69
	LSTM-FCN	97.07±0.10	79.38±0.63	81.48±2.13	80.52±1.45	80.93±0.58
	MP-ConvLSTM	**97.50±0.15**	**84.43±0.73**	82.16±2.26	89.81±1.98	**85.79±0.66**
USA	CVA	92.82	74.46	97.57	65.78	78.58
	PCA-CVA	92.97	74.93	**98.50**	65.88	78.96
	SVM	96.10±0.16	87.69±0.42	90.29±1.51	90.16±1.35	90.16±0.32
	GETNET	93.97±0.09	82.41±0.28	79.41±0.48	94.45±0.53	86.26±0.22
	PBCNN	95.72±0.26	86.80±0.69	88.32±2.37	90.55±1.99	89.43±0.52
	3D-CNN	96.31±0.14	88.25±0.38	91.54±1.64	89.77±1.56	90.59±0.29
	SSCNN-S	96.30±0.10	88.64±0.32	87.48±0.06	**95.06±0.62**	91.03±0.26
	SSA-SiamNet	95.80±0.41	87.41±1.06	86.71±2.91	93.37±1.49	90.01±0.79
	LSTM	95.07±0.11	85.20±0.42	85.66±1.43	90.72±1.99	88.18±0.36
	LSTM-FCN	93.58±0.17	81.09±0.41	79.10±1.79	92.25±1.32	85.15±0.29
	MP-ConvLSTM	**96.77±0.22**	**89.98±0.60**	91.26±1.81	92.79±1.42	**92.01±0.46**
Hermiston	SVM	98.41±0.03	93.45±0.10	77.11±0.46	77.27±0.41	77.19±0.21
	GETNET	98.98±0.06	95.42±0.11	90.93±0.73	97.73±0.05	93.99±0.40
	PBCNN	98.56±0.11	93.92±0.05	**96.23±0.54**	95.11±0.19	95.65±0.29
	3D-CNN	98.79±0.07	94.78±0.02	94.50±0.47	91.18±0.38	92.44±0.07
	SSCNN-S	99.02±0.06	95.76±0.24	92.06±1.43	97.64±0.20	94.60±0.81
	SSA-SiamNet	99.04±0.05	96.09±0.21	92.76±0.49	**97.92±0.09**	95.15±0.30
	LSTM	98.27±0.16	92.51±0.03	89.70±0.37	82.87 ±0.36	85.63±0.02
	LSTM-FCN	98.52±0.12	93.43±0.02	94.52±0.63	90.45±0.52	92.26±0.15
	MP-ConvLSTM	**99.19±0.08**	**96.53±0.10**	94.75±0.80	97.12±0.32	**95.85±0.47**

[1] The bold value indicates the most accurate result in each term.
[2] The CVA and PCA-CVA methods do not require training phase, and the standard deviation of all metrics is zero.

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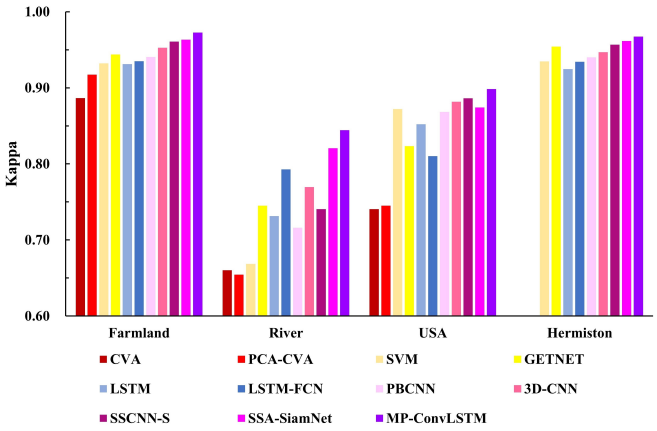

Fig. 9. Kappa coefficient comparisons in different methods for the four datasets.

1) Quantitative Evaluation: The OA accuracy of CVA and PCA-CVA was acceptable. However, the kappa coefficients were less accurate than the deep learning and SVM methods. Many false detections identified unchanged pixels as changed pixels by the two algorithms. The CVA method yielded the poorest detection results for the "Farmland" and "USA" datasets in kappa and F1, and the PCA-CVA method has the worst results for the "River" dataset in kappa and F1. For deep learning-based and SVM methods, they can be divided into four groups.

1) For SVM and GETNET model, only the spectral features were extracted for learning, and although the convolution structure was introduced in GETNET, it was for the extraction of the mixed affinity matrix reflecting the spectral information only.
2) PBCNN, 3-D-CNN, SSCNN-S, and SSA-SiamNet took both spatial and spectral features into account.
3) LSTM and LSTM-FCN considered the temporal–spectral features, and LSTM-FCN added 1-D convolution in the time dimension compared to LSTM.
4) Our proposed MP-ConvLSTM method took joint multiscale temporal–spatial–spectral features into account in feature extraction.

Due to the high proportion of the unchanged pixels in three datasets, the OA obtained by these four groups of models was relatively close, and the kappa coefficient is a better choice to evaluate the consistency of the CD map with the ground truth. Also, F1-score is a comprehensive indicator of the good precision and recall value.

The kappa coefficients and F1-scores of SVM, GETNET, LSTM, LSTM-FCN, and PBCNN were relatively close for the "Farmland" dataset, around 0.94 and 0.95, respectively. The models, such as 3-D-CNN, SSCNN-S, SSA-SiamNet, and MP-ConvLSTM, took the spatial–spectral features into account jointly and achieved higher kappa coefficients. Our proposed method outperformed the other SOTA models achieving the kappa coefficient of 0.9723 and the F1-score of 0.9800.

For the "River" dataset that was more difficult for CD, the performance of the compared models differed obviously in terms of kappa coefficient and F1-score. SVM yielded the lowest kappa coefficient and F1-score. At the same time, LSTM-FCN, SSA-SiamNet, and our proposed model all achieved better performances with both two criteria higher than 0.8, among which our proposed model again outperformed the others with the highest kappa coefficient of 0.8443 and the F1-score of 0.8579.

For the "USA" dataset, performances of the LSTM-FCN and GETNET ranked the poorest with kappa coefficient around 0.82 and F1-score around 0.86, and the LSTM also performed not satisfactorily, but SVM exhibited relatively better performance in terms of kappa coefficient about 0.877, which was close to PBCNN, 3-D-CNN, SSCNN-S, and SSA-SiamNet, as well as to our proposed model. The best performance was again ranked to our proposed model with the highest kappa coefficient of about 0.8998 and the F1-score of 0.9201.

For the "Hermiston" dataset, the ranks of CD accuracy for each method are rather close to those of the "Farmland" dataset. Notably, CNN has the highest precision but lower recall. Our proposed method has the highest OA, kappa, and F1.

From the overall evaluation of model performances, as shown in Fig. 9, it was apparent that the models only took spectral features (SVM and GETNET) or temporal–spectral features (LSTM and LSTM-FCN) into account yielded the relatively lower kappa coefficients than those considering spatial–spectral features according to the models' kappa coefficients of "Farmland," "USA," and "Hermiston" datasets. SSCNN-S, SSA-SiamNet, and our proposed model adopted the Siamese structure and outperformed 3-D-CNN and PBCNN on the three datasets. Since the LULC change type of "River" dataset between two timesteps is much more complex and this dataset includes more small and irregularly shaped areas, the spatial features are not significant enough compared to other datasets. Therefore, the outperformance of methods (PBCNN, 3-D-CNN, and SSCNN-S) that exploit the spatial–spectral features did not work well for the "River" dataset. The introduction of parallel LSTM structure with multiscale time-dependent modeling capability in MP-ConvLSTM can significantly improve the performances of the model in both binary and multiclass CD tasks.

In addition, for the CD task, the complexity of the model cannot be ignored. The wider and deeper the network, the longer time is required for calculations, but more fine-grained features can be captured [55]. The tradeoff between the model performance and the floating point of operations (Flops) was considered in our comparative experiments.

In Fig. 10 (the model complexity of GETNET far exceeds that of other deep learning models and will not be compared), the closer the position of the corresponding point of the model to the top-left corner of the figure is, the lower complexity and the higher accuracy of the model would be, and the better tradeoff between complexity and accuracy of the model was thus achieved. Experiments on three datasets of HSIs indicate that our proposed MP-ConvLSTM also outperformed the other

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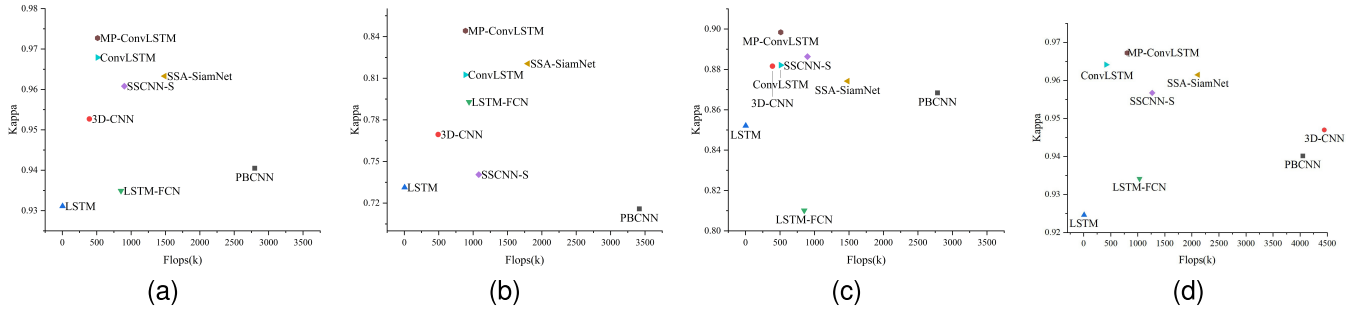

Fig. 10. Flops versus Kappa in different methods for the four datasets. (a) "Farmland." (b) "River." (c) "USA." (d) "Hermiston."

TABLE IV

TOTAL PARAMS AND FLOPS IN DIFFERENT DEEP LEARNING-BASED METHODS FOR THE FOUR DATASETS

Dataset	Method	Total Params(k)	Total Flops(k)
Farmland	GETNET	93415.258	186831.742
	PBCNN	210.402	2797.728
	3D-CNN	196.252	393.162
	SSCNN-S	18.068	902.124
	SSA-SiamNet	36.614	1486.056
	LSTM	98.272	3.104
	LSTM-FCN	424.050	853.197
	ConvLSTM	43.280	512.543
	MP-ConvLSTM	30.698	512.543
River	GETNET	154179.418	308360.062
	PBCNN	235.170	3416.928
	3D-CNN	245.788	492.234
	SSCNN-S	21.680	1082.724
	SSA-SiamNet	42.806	1795.656
	LSTM	159.590	3.964
	LSTM-FCN	468.082	941.261
	ConvLSTM	74.753	889.268
	MP-ConvLSTM	53.333	889.268
USA	GETNET	92182.362	184365.950
	PBCNN	209.826	2783.328
	3D-CNN	195.100	390.858
	SSCNN-S	17.984	897.924
	SSA-SiamNet	36.470	1478.856
	LSTM	97.022	3.084
	LSTM-FCN	423.026	851.149
	ConvLSTM	43.118	511.085
	MP-ConvLSTM	30.617	511.085
Hermiston	GETNET	232036.186	464073.598
	PBCNN	260.774	4050.784
	3D-CNN	296.6	4449.322
	SSCNN-S	25.428	1267.572
	SSA-SiamNet	49.21	2112.520
	LSTM	238.618	6.78
	LSTM-FCN	513.686	1032.461
	ConvLSTM	24.178	416.55
	MP-ConvLSTM	50.702	802.461

[1] For multi-class change detection, MP-ConvLSTM does not have the siamese structure for weight sharing, and the number of parameters is about twice that of ConvLSTM.

compared SOTA models in terms of the tradeoff between complexity and accuracy.

2) Qualitative Evaluation: For the "Farmland" dataset, as shown in Fig. 11, both CVA and PCA-CVA models presented apparent false detection at the boundary of crop fields. Most change maps obtained by the deep learning-based methods in the green box displayed varying degrees of false detection for small objects, and the CD results in the red box do not match well with the ground truth; however, the CD map obtained with our proposed model was more accurate compared with the others.

For the "River" dataset, as the CD maps for CVA and PCA-CVA shown in Fig. 12, it can be observed that many small objects were misidentified over the green box. The difference between the CD maps obtained in the red box was noticeable. However, the CD map for MP-ConvLSTM exhibited the best match with the ground truth.

For the "USA" dataset, land cover changes over the dataset, as shown in Fig. 13, were mainly circular irrigated farmland and riverbank in the lower part of the image, both CVA and PCA-CVA failed to identify some changed areas of the irrigated farmland over the green box. Moreover, the CD accuracy was poor for the riverbank as can be seen in the red box. Most CD maps obtained by the deep learning-based methods exhibited apparent differences in continuity within the green box compared with ground truth. CD map obtained by MP-ConvLSTM showed better consistency for both farmland and riparian changes in this regard.

For the "Hermiston" dataset as shown in Fig. 14, there are five land cover change types in the dataset. The differences between the CD maps of the different methods were mainly found in change type "c1" in the white box, and the other methods were not as accurate as MP-ConvLSTM.

IV. DISCUSSION

A. Ablation Study of MP-ConvLSTM

In the proposed MP-ConvLSTM, the parallel architecture was constructed to extract and learn multilevel spectral–spatial time-dependent features by combining hidden states from different paths (as shown in Figs. 2, 5, and 6) to improve the performance of the model in CD.

To demonstrate the effectiveness and reliability of the parallel architecture, we trained both MP-ConvLSTM (with parallel architecture) and stacking ConvLSTM (without parallel architecture) with the rest of the structure remaining the same to investigate the difference of their performances for CD tasks. As shown in Table V, the F1 and kappa coefficients obtained from those three datasets were significantly higher for the proposed MP-ConvLSTM than those for stacking ConvLSTM. Under the condition that the number of floating point of operations was identical for both models, the accuracy of CD for the MP-ConvLSTM was dramatically improved, and for the "Farmland" dataset, the F1 and kappa coefficient improved by about 0.39% and 0.47%, respectively. For the "River" dataset with the lowest proportion of training samples and

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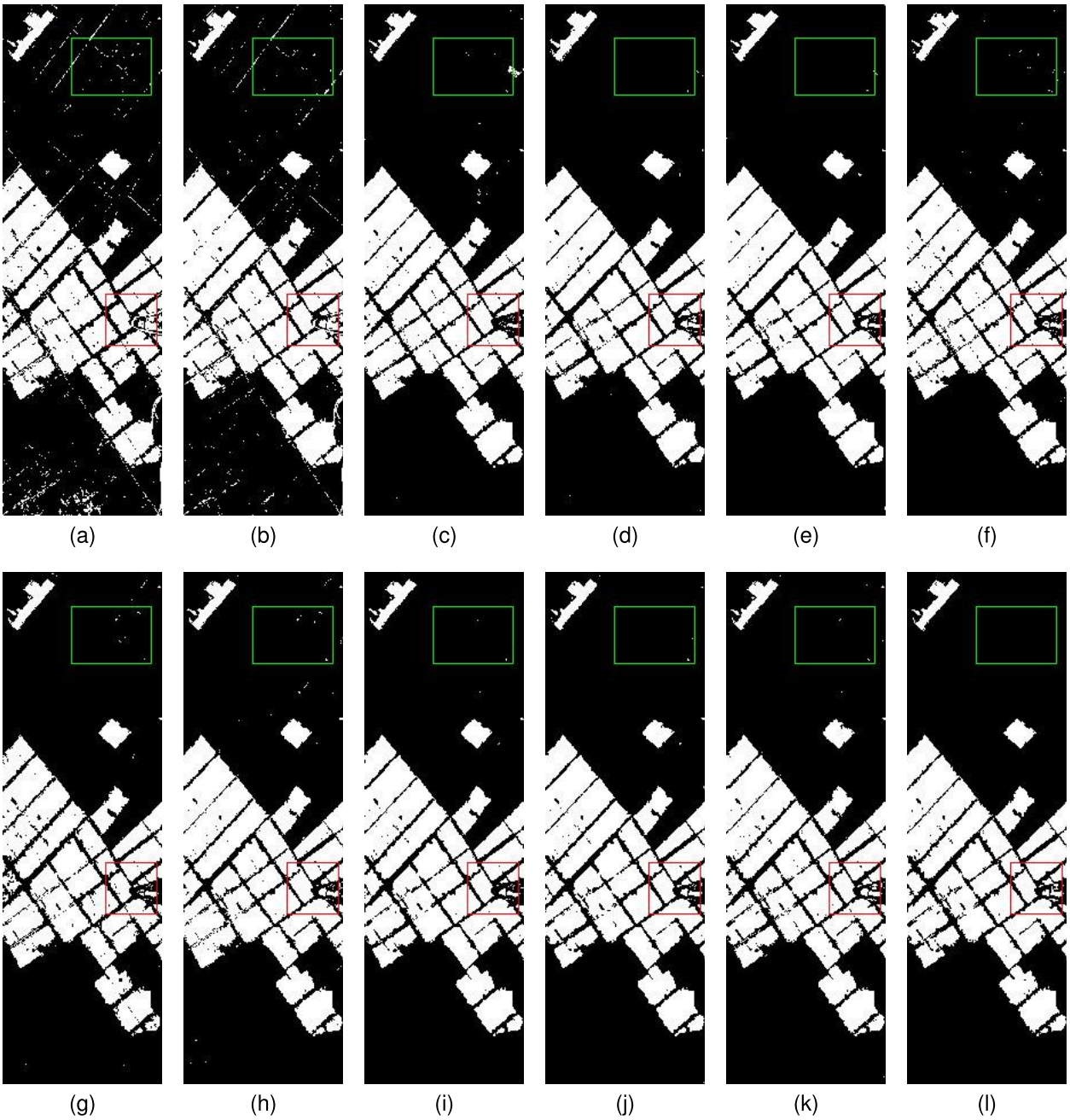

Fig. 11. CD maps for the "Farmland" dataset. (a) CVA. (b) PCA-CVA. (c) SVM. (d) PBCNN. (e) 3-D-CNN. (f) LSTM. (g) LSTM-FCN. (h) GETNET. (i) SSCNN-S. (j) SSA-SiamNet. (k) MP-ConvLSTM. (l) Ground truth.

the most complex LULC types, the significant improvement was obtained with our proposed model, and the F1 and kappa coefficient improved by about 2.94% and 3.14%, respectively. For the "USA" dataset, the F1 and kappa coefficient improved by about 1.60% and 1.95%, respectively. For the "Hermiston" dataset, the F1 and kappa coefficient improved by about 0.21% and 0.19%, respectively. These results implied that the lower the proportion of the training dataset, the more significant the improvement might be achieved by the parallel architecture.

B. Impact of Dimension Reduction

Since the HSI bands are usually huge, direct input to the ConvLSTM block will result in a considerable increase of network parameters, which makes training convergence of the model difficult. PCA and independent component analysis (ICA) are commonly used hyperspectral dimensionality reduction methods, but PCA reduces hyperspectral dimensionality with the expense of partial useful information lost. ICA can decompose hyperspectral into independent signals but cannot exclude noisy signals. Feature extraction and dimensionality reduction can be performed on image pairs simultaneously by the Siamese CNN, and the number of convolutional kernels determines the number of channels of the feature map after dimensionality reduction. In the present study, we set the reduction ratio to 16, and the number of channels of the input to multipath ConvLSTM block is reduced to 1/16 of the original input image's channel.

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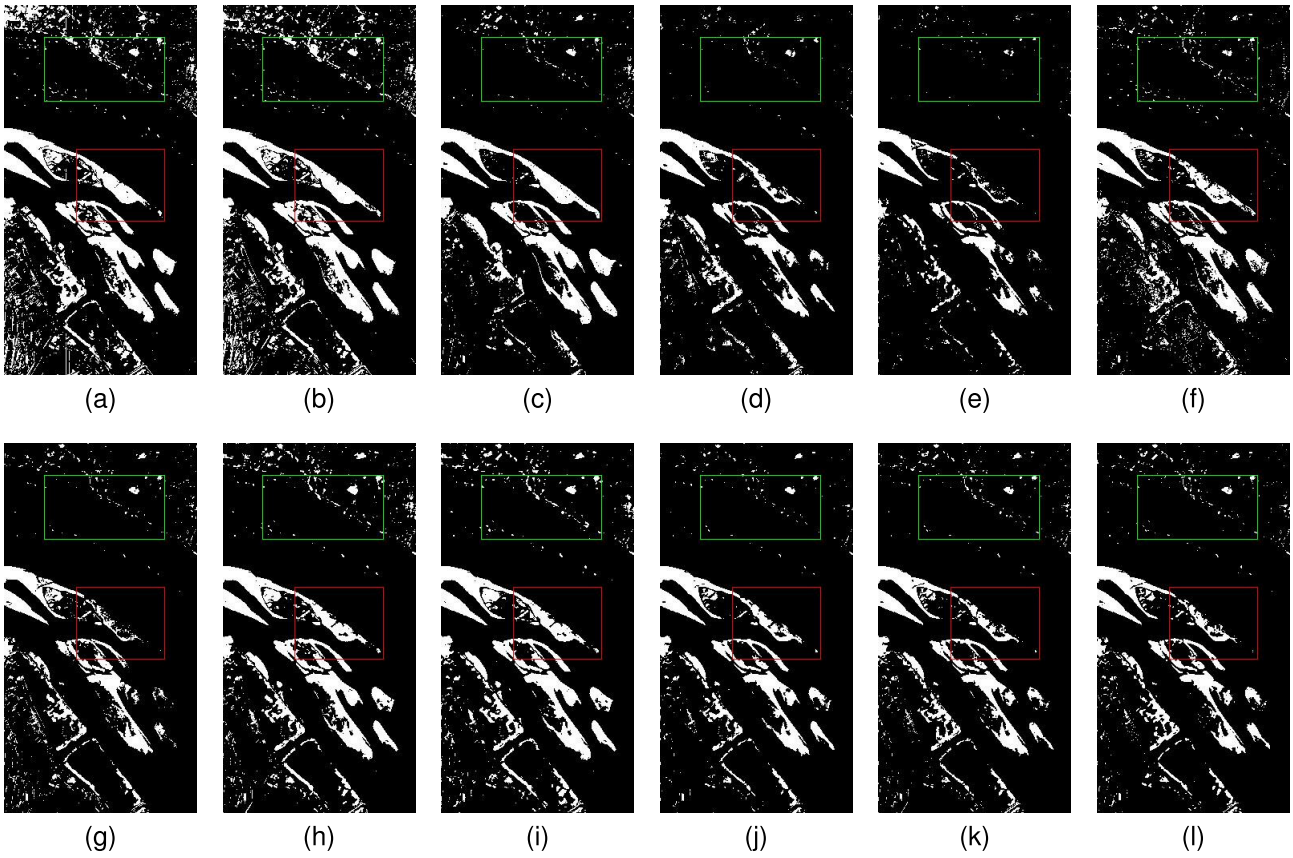

Fig. 12. CD maps for the "River" dataset. (a) CVA. (b) PCA-CVA. (c) SVM. (d) PBCNN. (e) 3-D-CNN. (f) LSTM. (g) LSTM-FCN. (h) GETNET. (i) SSCNN-S. (j) SSA-SiamNet. (k) MP-ConvLSTM. (l) Ground truth.

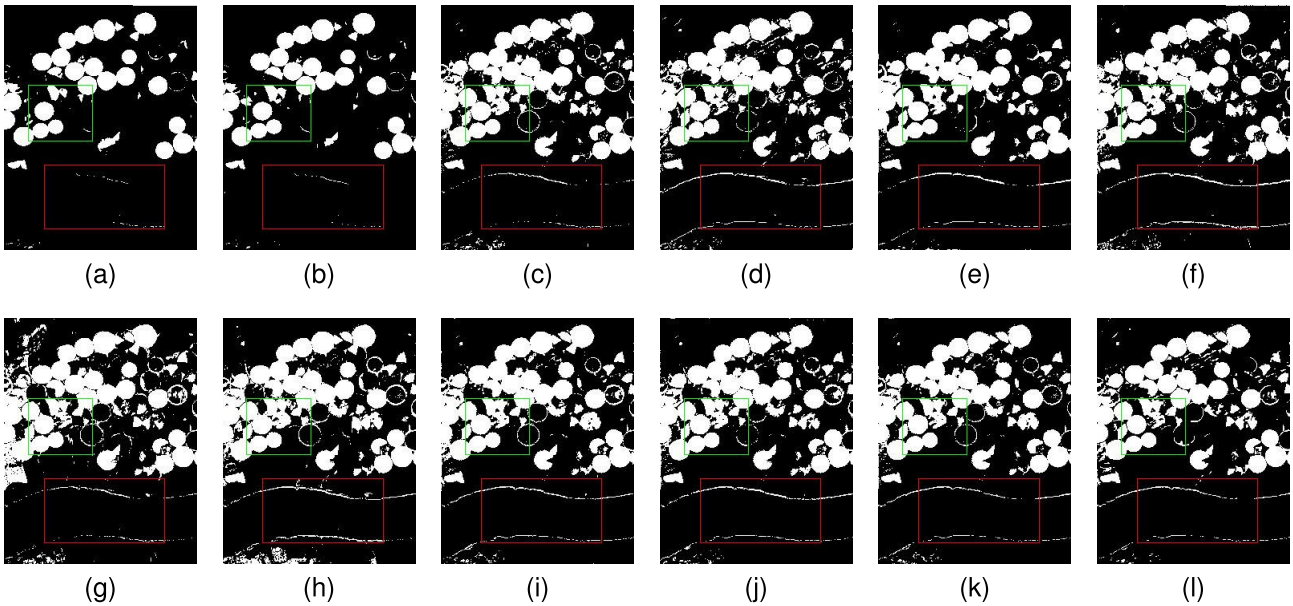

Fig. 13. CD maps for the "USA" dataset. (a) CVA. (b) PCA-CVA. (c) SVM. (d) PBCNN. (e) 3-D-CNN. (f) LSTM. (g) LSTM-FCN. (h) GETNET. (i) SSCNN-S. (j) SSA-SiamNet. (k) MP-ConvLSTM. (l) Ground truth.

1) Reduction Method: As presented in Table VI, for binary CD, the Siamese CNN outperformed the PCA and ICA by its nonlinear modeling capability of reducing hyperspectral dimensionality while mining deep features, and both the F1 and kappa coefficients in CD tasks get improved for the MP-ConvLSTM with three binary CD HSIs. The Siamese structure can highlight differences but is inadequate for discriminating complex types of change and performs poorly in

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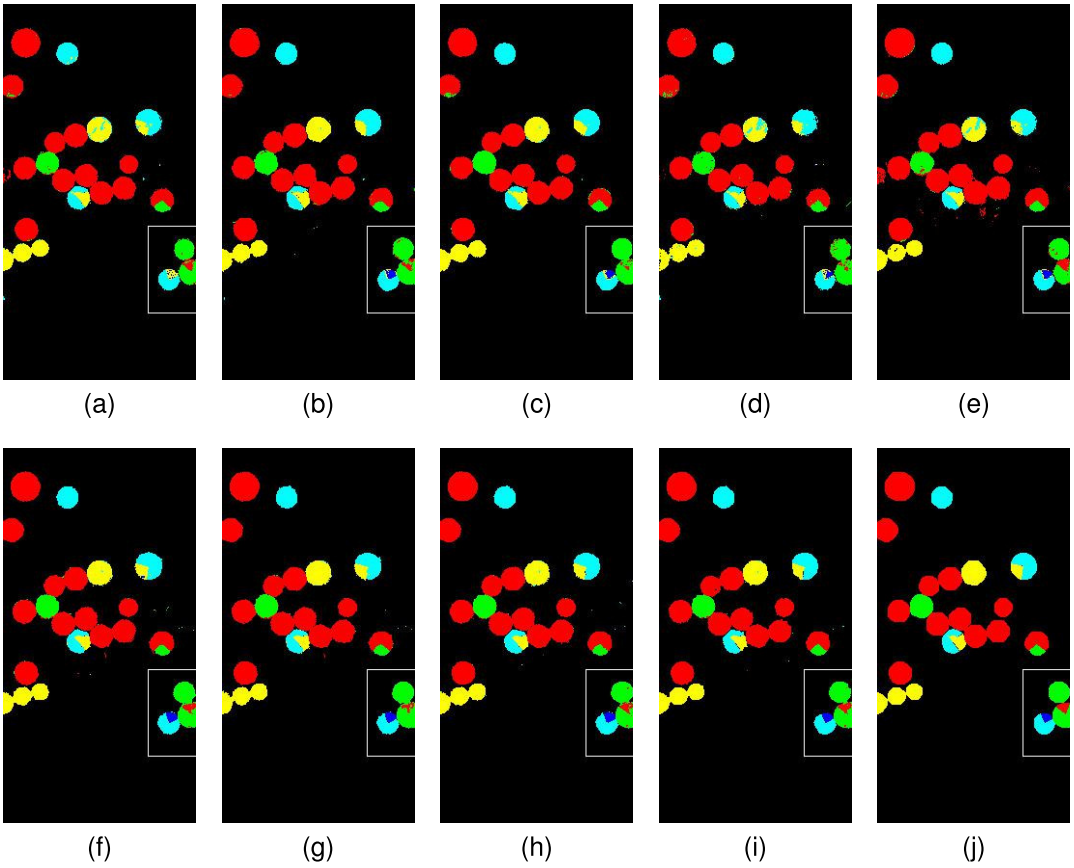

Fig. 14. CD maps for the "Hermiston" dataset. (a) SVM. (b) PBCNN. (c) 3-D-CNN. (d) LSTM. (e) LSTM-FCN. (f) GETNET. (g) SSCNN-S. (h) SSA-SiamNet. (i) MP-ConvLSTM. (j) Ground truth.

TABLE V

ACCURACY OF THE CONVLSTM AND MP-CONVLSTM FOR THE FOUR DATASETS

Dataset	Architecture	OA(%)	Kappa(%)	Precision(%)	Recall(%)	F1(%)
Farmland	Non-parallel	98.68±0.05	96.76±0.19	97.34±0.27	97.91±0.14	97.61±0.07
	Parallel	**98.89±0.06**	**97.23±0.15**	**97.80±0.18**	98.20±0.27	**98.00±0.11**
River	Non-parallel	97.31±0.22	81.29±0.68	81.92±1.39	83.86±1.26	82.85±0.53
	Parallel	**97.50±0.15**	**84.43±0.73**	82.16±2.26	89.81±1.98	**85.79±0.66**
USA	Non-parallel	96.19±0.24	88.03±0.39	**91.17±1.76**	89.63±0.83	90.40±0.28
	Parallel	**96.77±0.22**	**89.98±0.60**	91.26±1.81	92.79±1.42	**92.01±0.46**
Hermiston	Non-parallel	99.13±0.05	96.34±0.16	94.61±0.53	96.72±0.53	95.64±0.41
	Parallel	**99.19±0.08**	**96.53±0.10**	**94.75±0.80**	**97.12±0.32**	**95.85±0.47**

TABLE VI

ACCURACY OF DIFFERENT REDUCTION METHOD FOR THE FOUR DATASETS

Dataset	Method	OA(%)	Kappa(%)	Precision(%)	Recall(%)	F1(%)
Farmland	PCA	98.22±0.01	95.52±0.15	96.32±0.35	97.33±0.37	96.77±0.11
	ICA	98.22±0.02	95.60±0.15	96.39±0.29	97.24±0.22	96.79±0.14
	CNN	98.76±0.01	96.92±0.27	97.80±0.50	97.67±0.51	97.72±0.21
	Siam CNN	**98.89±0.06**	**97.23±0.15**	**97.80±0.18**	**98.20±0.27**	**98.00±0.11**
River	PCA	95.81±0.11	72.52±0.70	69.23±0.72	81.59±0.92	74.94±0.65
	ICA	95.89±0.06	70.57±0.46	76.23±1.68	69.47±2.00	72.74±0.44
	CNN	97.36±0.12	82.69±0.81	79.26±0.67	**89.86±1.12**	84.14±0.74
	Siam CNN	**97.50±0.15**	**84.43±0.73**	**82.16±2.26**	89.81±1.98	**85.79±0.66**
USA	PCA	95.92±0.09	87.47±0.36	88.60±0.89	91.37±1.42	89.95±0.31
	ICA	95.67±0.15	86.54±0.34	87.84±1.56	90.82 ±1.18	89.32±0.24
	CNN	96.73±0.10	89.65±0.36	**90.30±0.57**	**93.59±1.02**	91.80±0.30
	Siam CNN	**96.77±0.22**	**89.98±0.60**	91.26±1.81	92.79±1.42	**92.01±0.46**
Hermiston	PCA	**99.19±0.08**	**96.53±0.10**	**94.75±0.80**	97.12±0.32	**95.85±0.47**
	ICA	99.13±0.02	96.51±0.21	94.26±0.46	97.03±0.30	95.61±0.23
	CNN	98.83±0.03	94.92±0.11	92.92±0.67	**97.17±0.45**	94.89±0.30
	Siam CNN	98.88±0.03	95.23±0.20	93.01±0.63	96.49±0.44	94.62±0.35

multiclass CD compared to traditional dimensionality reduction methods.

For the "Farmland" dataset, the F1 and kappa coefficients were about 0.9800 and 0.9723, respectively. For the "River" dataset that contains the most complicated LULC change types, the F1 and kappa coefficients were 0.8579 and 0.8443, respectively, compared to the nonweighted sharing CNN, which improved about 1.65% and 1.74%, respectively. The F1 and kappa coefficients for the "USA" dataset were 0.9201 and 0.8998, respectively. The Siamese CNN used in our proposed MP-ConvLSTM only had one convolutional layer, and the Siamese structure made the network architecture compacted.

2) Reduction Ratio: The effects of different dimensionality reduction ratios on model performance were investigated, and the results are given in Table VII. For the "Farmland" dataset, both F1 and kappa coefficients reached the maximum when the reduction ratio was set to 8. For the "River" dataset, both F1 and kappa coefficients become the maximum when the reduction ratio is 16. For the "USA" dataset, the OA was close to about 0.967 when the reduction ratio became 8, 12, and 16. At the same time, the F1 and kappa coefficient reached the maximum at a reduction ratio of 16. For the "Hermiston" dataset, F1 reached the most when the reduction ratio was set to 8, kappa reached the maximum when the reduction ratio was set to 12.

Comparing the F1 and kappa coefficients of the three binary CD datasets' experimental results under different reduction ratio conditions, we found that datasets, including a different

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TABLE VII
ACCURACY OF DIFFERENT REDUCTION RATIOS FOR THE FOUR DATASETS

Dataset	Ratio	OA(%)	Kappa(%)	Precision(%)	Recall(%)	F1(%)
Farmland	8	**98.94±0.01**	**97.30±0.12**	**98.26±0.12**	97.85±0.24	**98.04±0.11**
	12	98.92±0.02	97.29±0.25	98.03±0.32	98.08±0.38	98.02±0.23
	16	98.89±0.06	97.23±0.15	97.80±0.18	**98.20±0.27**	98.00±0.11
	20	98.80±0.02	97.18±0.16	98.15±0.20	97.69±0.29	97.92±0.15
River	8	97.57±0.14	84.39±0.82	83.38±1.81	87.94±1.22	85.65±0.74
	12	97.48±0.08	83.34±0.52	**83.64±1.46**	85.50±1.28	84.56±0.48
	16	97.50±0.15	**84.43±0.73**	82.16±2.26	**89.81±1.98**	**85.79±0.66**
	20	**97.58±0.06**	83.58±0.56	83.50±1.95	86.04±2.48	84.75±0.54
USA	8	96.72±0.12	89.65±0.28	**92.70±1.52**	90.79±1.34	91.74±0.20
	12	96.73±0.16	89.82±0.40	91.81±1.67	92.15±1.48	91.88±0.30
	16	**96.77±0.22**	**89.98±0.60**	91.26±1.81	92.79±1.42	**92.01±0.46**
	20	96.60±0.09	89.46±0.27	89.50±1.38	**93.87±1.60**	91.62±0.23
Hermiston	8	99.18±0.04	**96.58±0.28**	**95.49±0.28**	96.67±0.22	**96.07±0.35**
	12	99.18±0.07	96.43±0.20	93.98±0.56	**97.59±0.47**	95.71±0.37
	16	**99.19±0.08**	96.53±0.10	94.75±0.80	97.12±0.32	95.85±0.47
	20	99.06±0.05	96.05±0.22	94.14±0.54	97.00±0.44	95.52±0.29

TABLE VIII
ACCURACY OF DIFFERENT PATCH SIZES INPUT FOR THE FOUR DATASETS

Dataset	Size	OA(%)	Kappa(%)	Precision(%)	Recall(%)	F1(%)
Farmland	3	**98.89±0.06**	**97.23±0.15**	**97.80±0.18**	**98.20±0.27**	**98.00±0.11**
	5	98.80±0.02	97.02±0.05	97.69±0.11	97.74±0.14	97.71±0.04
	7	98.71±0.09	96.85±0.23	97.71±0.09	97.68±0.35	97.65±0.16
River	3	**97.50±0.15**	**84.43±0.73**	82.16±2.26	89.81±1.98	**85.79±0.66**
	5	97.34±0.07	82.41±0.34	79.37±1.51	**88.92±1.55**	83.83±0.30
	7	96.28±0.08	71.02±0.41	**84.91±1.93**	63.96±2.34	72.98±0.38
USA	3	**96.77±0.22**	**89.98±0.60**	**91.26±1.81**	92.79±1.42	**92.01±0.46**
	5	96.46±0.12	89.17±0.40	88.96±1.44	**94.13±1.72**	91.39±0.33
	7	96.28±0.10	88.69±0.37	89.67±1.28	91.81±1.23	90.72±0.31
Hermiston	3	**99.19±0.08**	**96.53±0.10**	**94.75±0.80**	97.12±0.32	**95.85±0.47**
	5	99.17±0.03	96.52±0.22	94.32±0.43	97.31±0.28	95.71±0.29
	7	99.12±0.02	96.41±0.27	94.11±0.57	**97.52±0.68**	95.74±0.36

number of bands, have different sensitivity-to-reduction ratios. The "Farmland" and "USA" datasets used in this study were removed more noise bands and retain 155 and 154 bands, respectively. These two datasets are relatively insensitive to the reduction ratio and the four reduction rations have less effect on the CD accuracy, while the "River" dataset retains 198 bands, which contains more noise bands, and the CD results of this dataset are more sensitive to the reduction ratio.

C. Impact of Patch Size

The effects of patch size on the performances of the models for CD were also quantitatively evaluated. As listed in Table VIII, it was noted that the best performances were obtained for the patch size of 3×3 as the model input. The accuracy of the CD map obtained tended to decrease as the patch size increased. The smaller the patch size as input was, the less complex of the model was needed. Input with a small patch size could effectively reduce the high requirement on the memory of the computer during the model training and inference phases.

D. Impact of ECA Module

The attention mechanism of efficient channel attention for deep convolutional neural networks (ECANet) was adopted to aggregate the learned joint multiscale temporal–spatial–spectral features. To verify the effect of the ECA module on the performance of our proposed model, the impact of different attention mechanisms, such as SENet and non-local neural network (NLNet) [56], was introduced to our proposed model for comparisons; meanwhile, the case without the attention mechanism was also considered. As shown in Table IX, the ECA module dramatically improved the performances of our model for the CD with the "River" and "Hermiston" dataset, while the model with SE or non-local (NL) attention mechanism performed not as well as ECA. For the "USA" dataset, both ECA and NL attention mechanisms contributed to the increase of accuracy. However, the SE attention mechanism performed not well, and most metrics were worse than the model without an attention mechanism. For the "Farmland" dataset, the F1 and kappa coefficients were also improved by attention mechanism, but no matter what attention mechanism was used, the improvements of all metrics were almost the same.

TABLE IX
ACCURACY OF DIFFERENT ATTENTION MECHANISMS FOR THE FOUR DATASETS

Dataset	Attention	OA(%)	Kappa(%)	Precision (%)	Recall(%)	F1(%)
Farmland	ECA	98.86±0.06	**97.23±0.15**	97.80±0.18	98.20±0.27	98.00±0.11
	SE	98.83±0.04	97.21±0.14	97.69±0.26	**98.36±0.32**	**98.02±0.23**
	NL	**98.89±0.02**	97.17±0.15	97.85±0.39	98.13±0.44	97.98±0.14
	No Attention	98.80±0.10	97.03±0.24	**97.86±0.04**	97.78±0.31	97.86±0.18
River	ECA	**97.50±0.15**	**84.43±0.73**	83.96±2.26	**89.81±1.98**	**85.79±0.66**
	SE	97.28±0.10	81.28±0.46	82.62±1.31	82.54±2.10	82.57±0.37
	NL	97.63±0.07	83.73±0.34	83.83±1.54	86.59±2.05	85.03±0.32
	No Attention	96.40±0.08	71.04±0.47	**90.87±1.29**	60.83±1.12	72.92±0.44
USA	ECA	96.67±0.22	89.78±0.60	**91.26±1.81**	92.79±1.42	**91.81±0.46**
	SE	96.39±0.11	88.74±0.33	89.59±0.56	92.86±0.64	91.15±0.27
	NL	**96.69±0.09**	**89.81±0.25**	90.23±0.58	**93.48±0.32**	91.72±0.19
	No Attention	96.55±0.15	89.33±0.41	90.68±0.81	92.20±0.24	91.44±0.32
Hermiston	ECA	**99.19±0.08**	**96.53±0.10**	94.75±0.80	**97.12±0.32**	**95.85±0.47**
	SE	99.15±0.15	96.47±0.23	94.01±0.28	95.52±0.38	94.63±0.27
	NL	99.17±0.09	96.50±0.10	**95.13±0.35**	96.35±0.39	95.29±0.35
	No Attention	99.12±0.19	96.41±0.17	93.78±0.13	95.93±0.33	94.72±0.21

V. CONCLUSION

In the present study, a multipath ConvLSTM neural network, entitled MP-ConvLSTM, was proposed. The network takes advantage of LSTM and CNN through the skillfully designed parallel architecture that enables the DLN to learn multilevel temporal dependencies of bitemporal HSIs. Also, for extracting joint multiscale temporal–spatial–spectral features, hidden states from different paths of ConvLSTM are combined, which enhances robustness to pseudo-changes resulted from the changes of HSI acquisition conditions and various noises and strengthens the capacity of multiscale representation. In the proposed MP-ConvLSTM, the ECA module is introduced to refine the most representative features of different paths, and the Siamese CNN is adopted for reducing the dimensionality of HSIs and extracting preliminary features to build up an end-to-end trainable model for the CD with HSIs.

The effectiveness and reliability of the proposed MP-ConvLSTM for binary and multiclass CD were investigated using four different HSIs benchmark datasets. The results of the CD map were evaluated visually and statistically by different accuracy indices. In addition, the CD results of our proposed method were compared with those obtained from several SOTA CD algorithms. The results confirmed that our proposed MP-ConvLSTM was superior to other compared SOTA models for CD and had a better tradeoff between complexity and accuracy.

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Nevertheless, the feasibility of the proposed network architecture should be explored in applications with various types of remotely sensed data for CD and temporal–spatial data modeling in the near future.

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AUTHOR CONTRIBUTIONS

Changjiang Shi, Zhijie Zhang, and Wanchang Zhang conceived and designed the experiment. Changjiang Shi and Zhijie Zhang performed the experiments and analyzed the data. Changjiang Shi wrote the first version of the manuscript. Zhijie Zhang and Wanchang Zhang revised the manuscript to improve the quality of the work. All authors have read the final manuscript. Prof. Chuanrong Zhang and Prof. Qiang Xu gave many constructive suggestions and helped the final edition of this article.

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